LAB 01

Cloud Computing

Lab Number & Lab Title:Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

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Department/Section:-Software Engineering/Section A

Course Code:-CC

GitHub Username: Eman-Tanveer123

Task 01:-Create and Configure Your GitHub Account:-

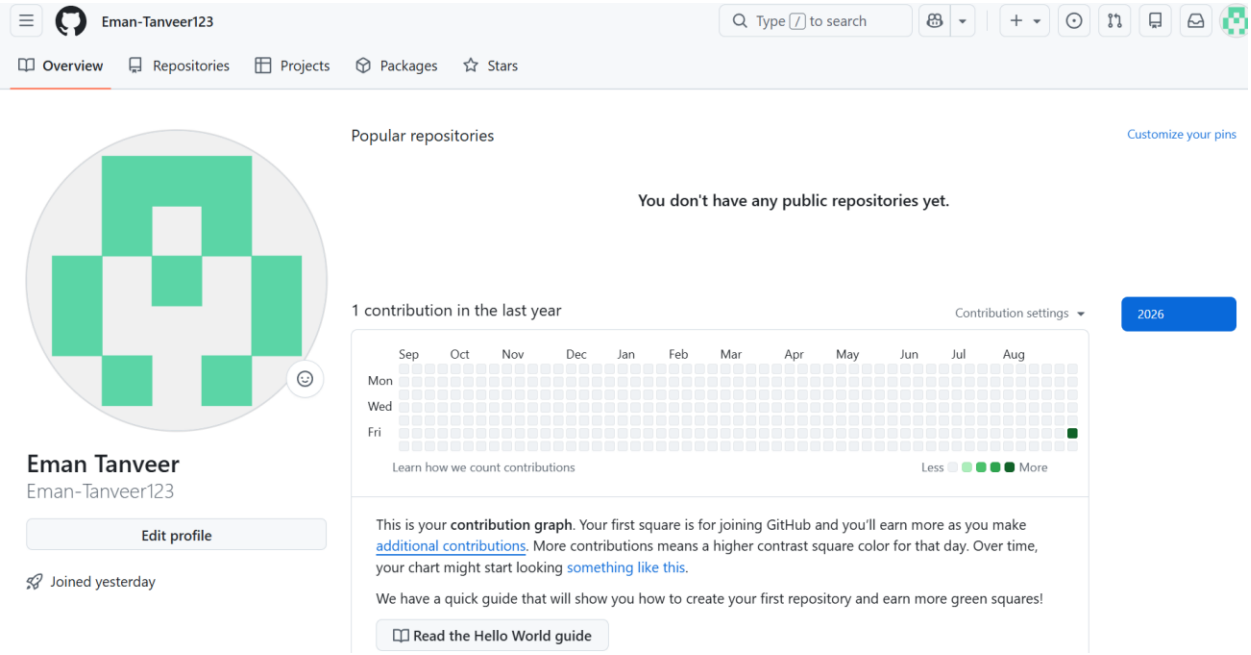


Figure 1.1:github_profile.png

Caption:-Overview of personal GitHub profile showing username and account details.

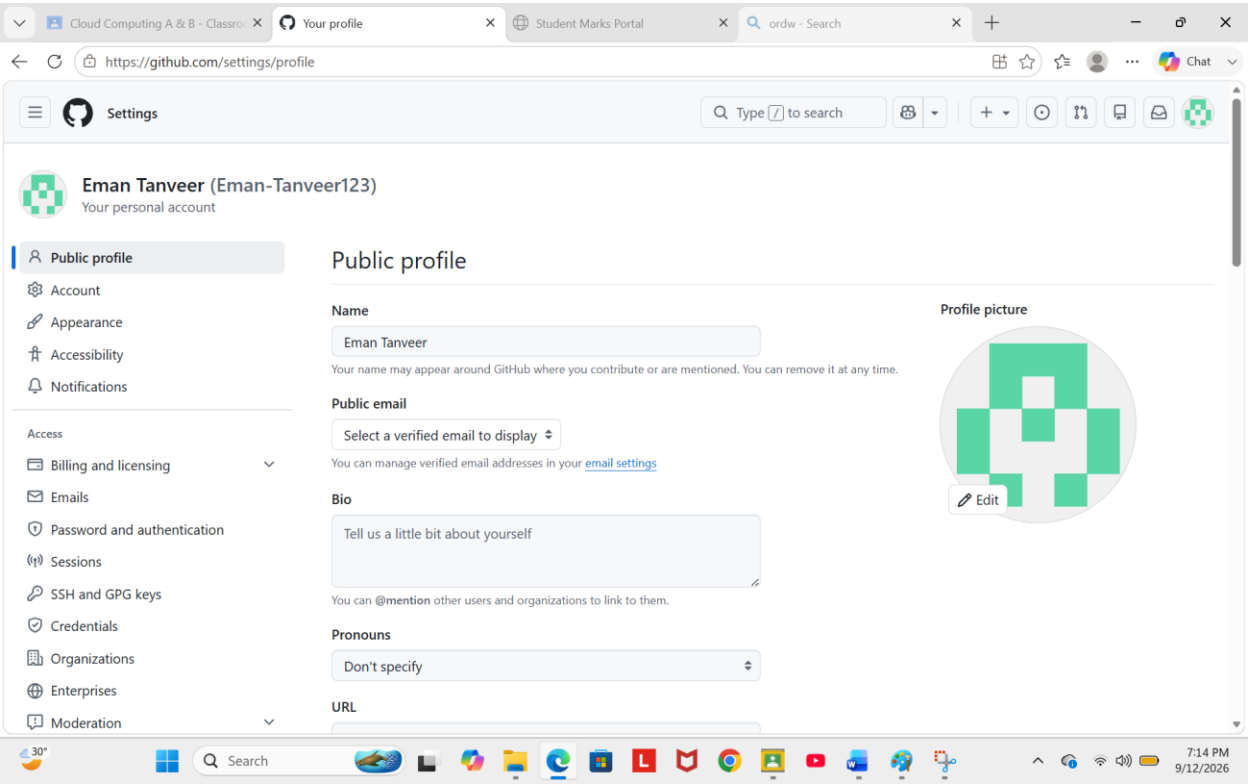


Figure 1.2:github_actual_name.png

Caption:-Verification of complete actual name configured in GitHub Public Profile settings.

Task 2:-Register in the Student Marks Portal:-

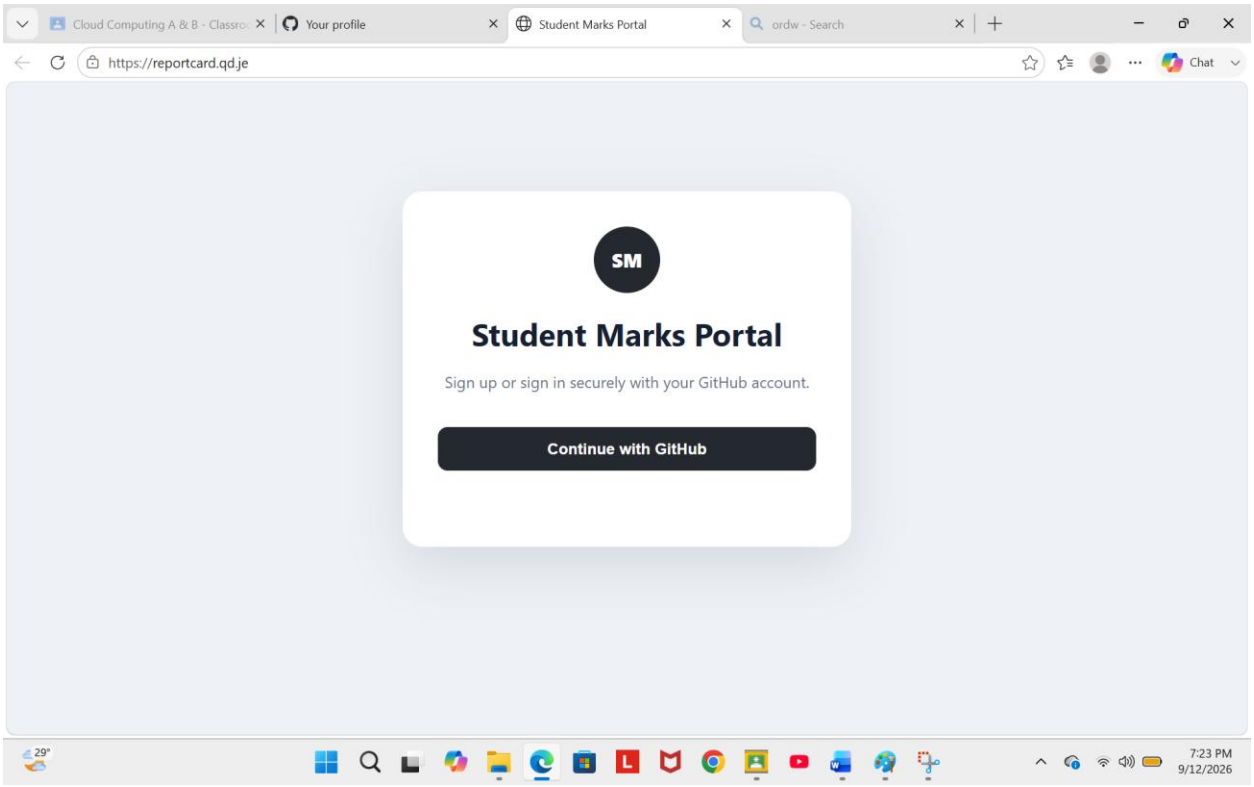


Figure 2.1:portal_login_page.png

Caption:-Main login page of the Student Marks Portal.

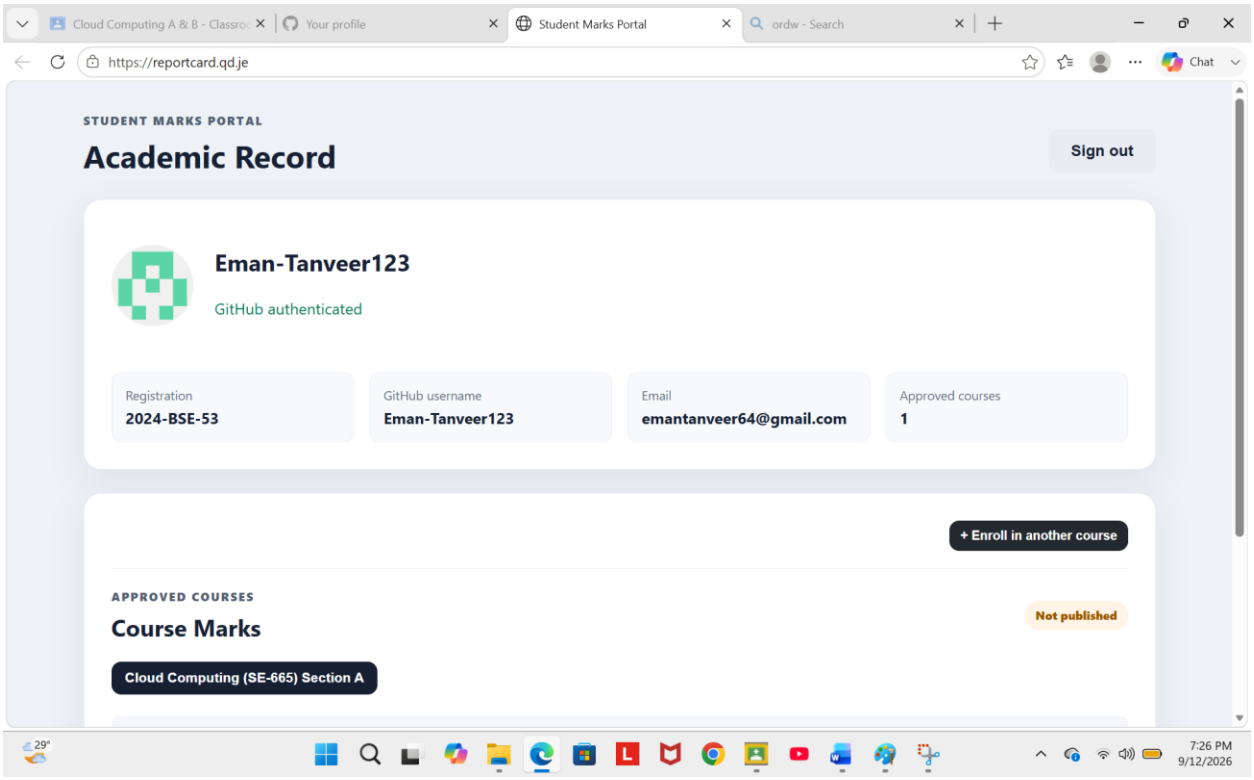


Figure 2.2:portal_github_profile.png

Caption:-GitHub profile integration inside the Student Marks Portal.

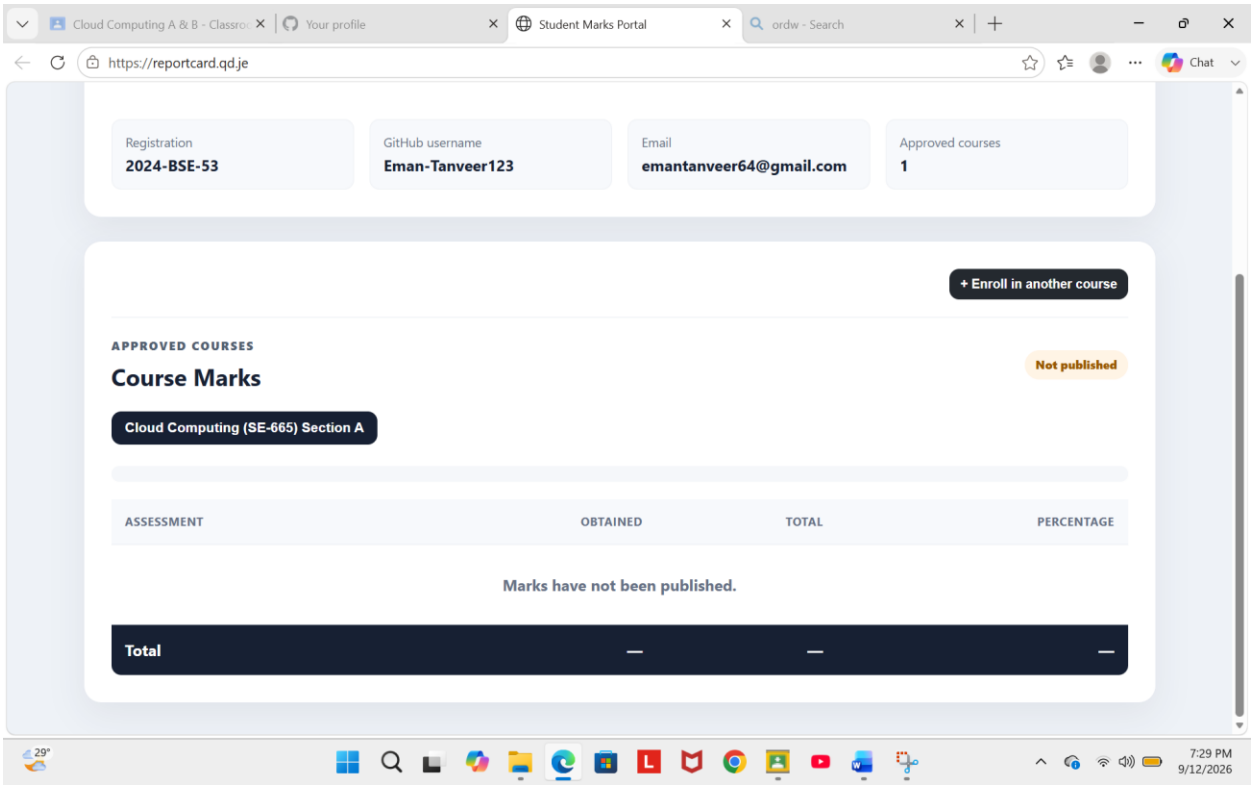


Figure 2.3:portal_enrollment_submitted.png

Caption:- Submitted student enrollment status showing registration number and course details.

Task 3:-Download VMware Workstation Pro and Ubuntu Server:

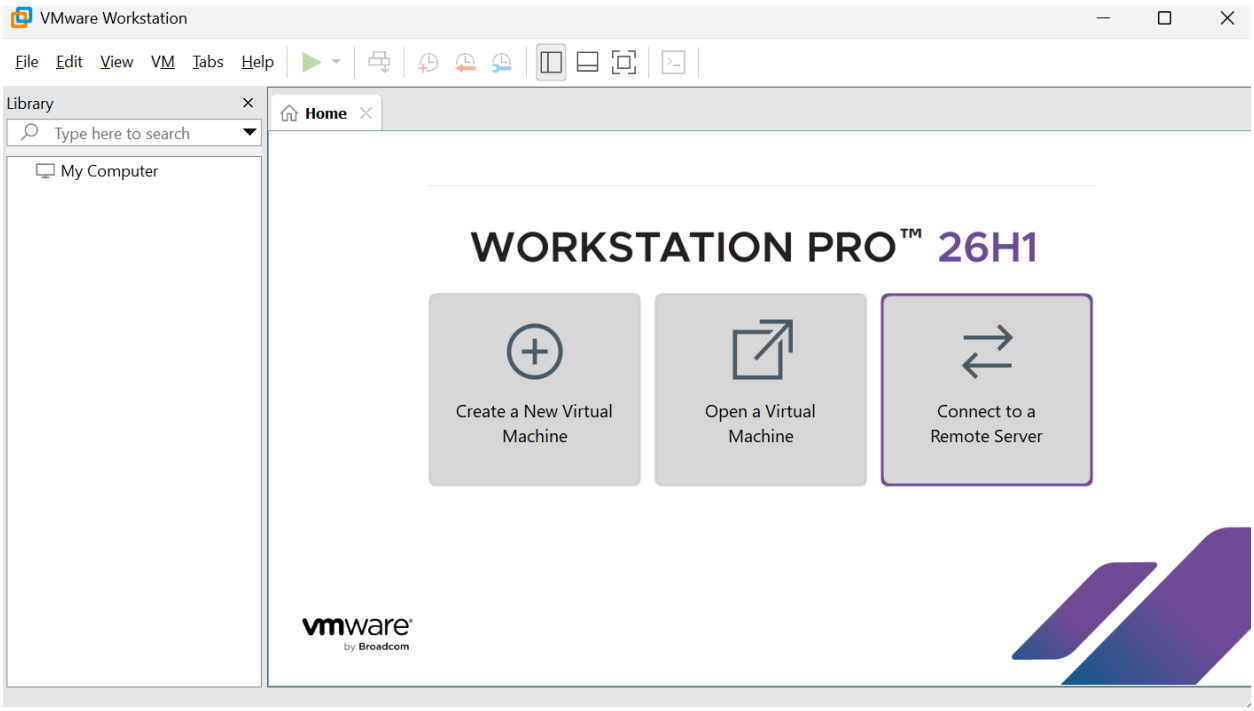


Figure 3.1:vmware_workstation.png

Caption:-Verification of VMware Workstation Pro running on Windows.

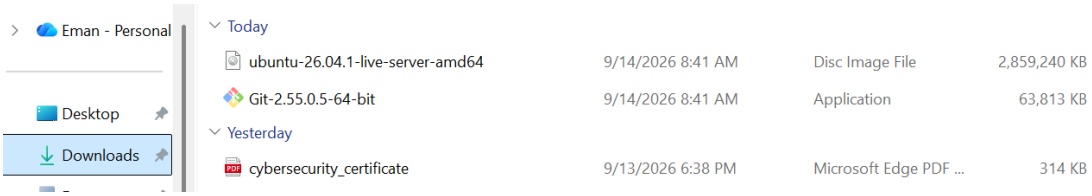


Figure 3.2:ubuntu_server_iso.png

Caption:-Downloaded Ubuntu Server LTS ISO image file.

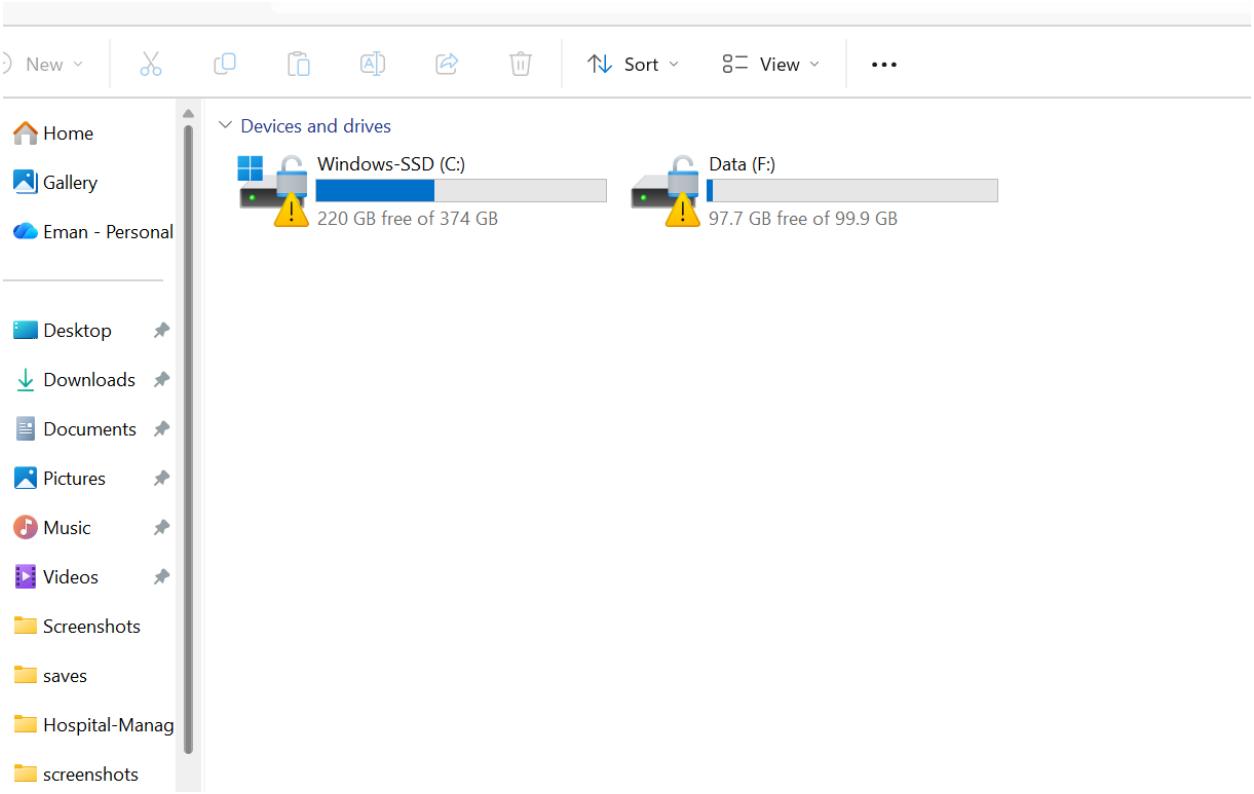


Figure 3.3:available_storage.png

Caption:-Available system disk storage verification for virtual machine creation.

Task 4:-Create the Ubuntu Server Virtual Machine:

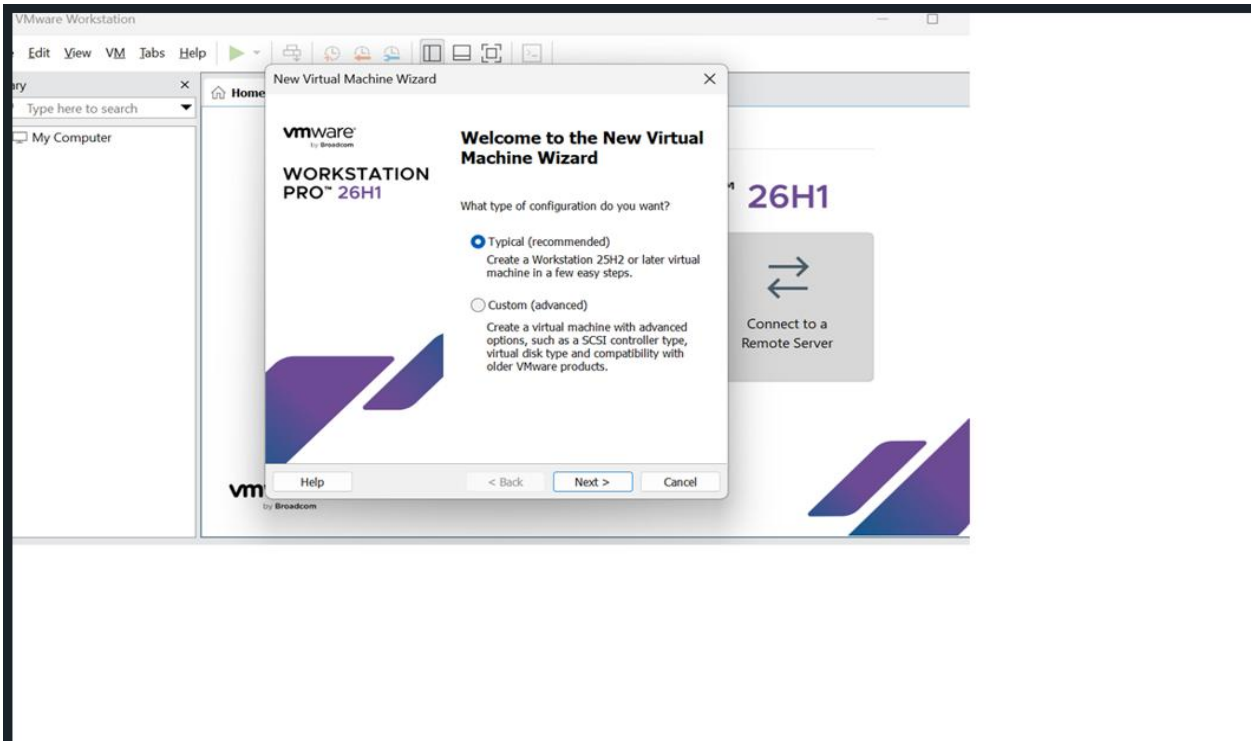


Figure 4.1:vm_creation_wizard.png

Caption:-Initial step of VMware Virtual Machine creation wizard.

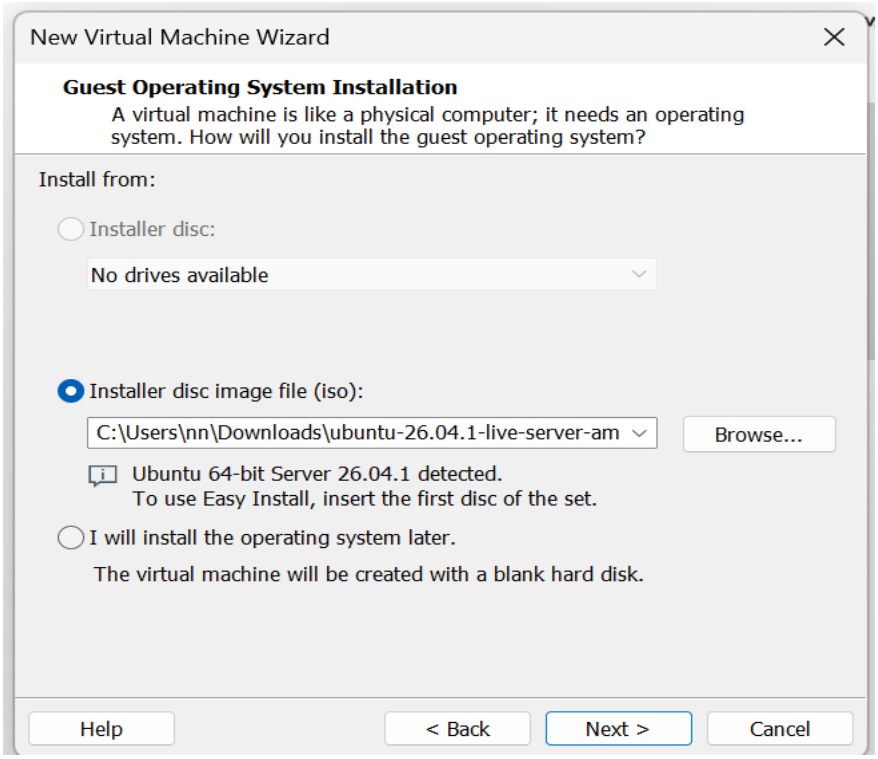


Figure 4.2:vm_typical_configuration.png

Caption:-Selecting Typical virtual machine configuration.

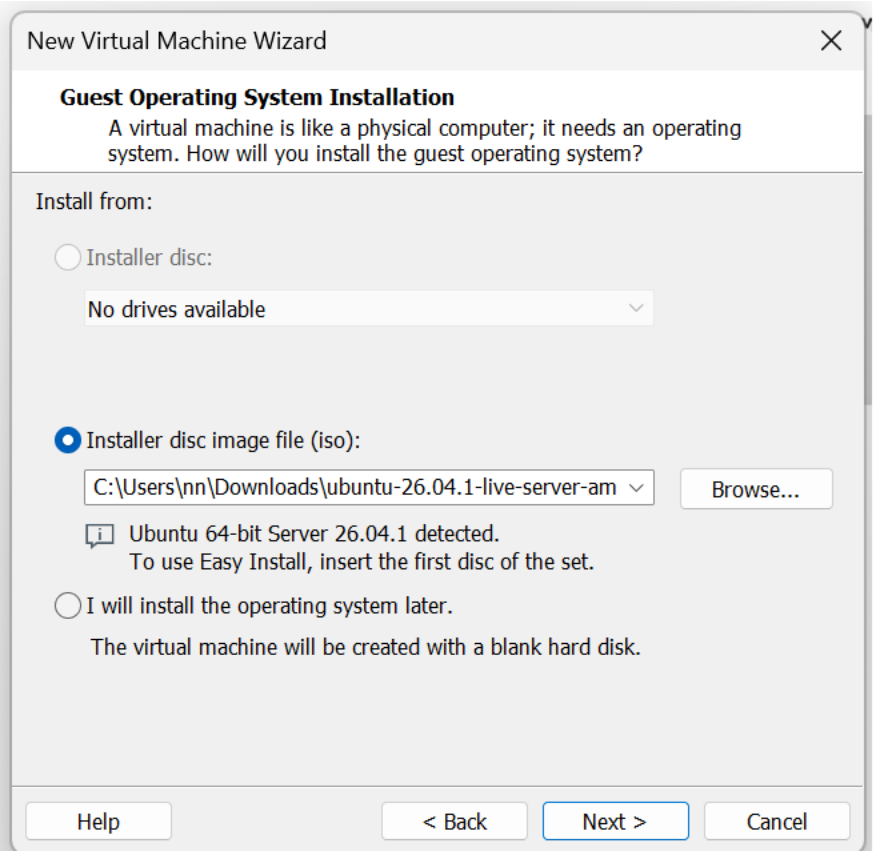


Figure 4.3:vm_ubuntu_iso_selected.png

Caption:-Selecting the downloaded Ubuntu Server ISO image path.

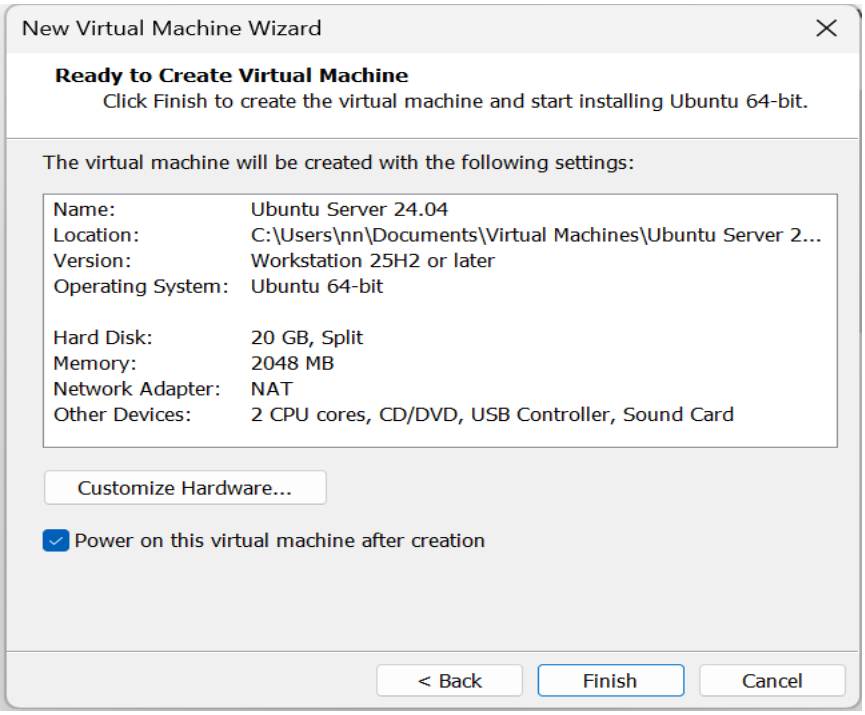


Figure 4.4: vm_configuration_summary.png

Caption:-Final hardware configuration summary for Ubuntu Server VM.

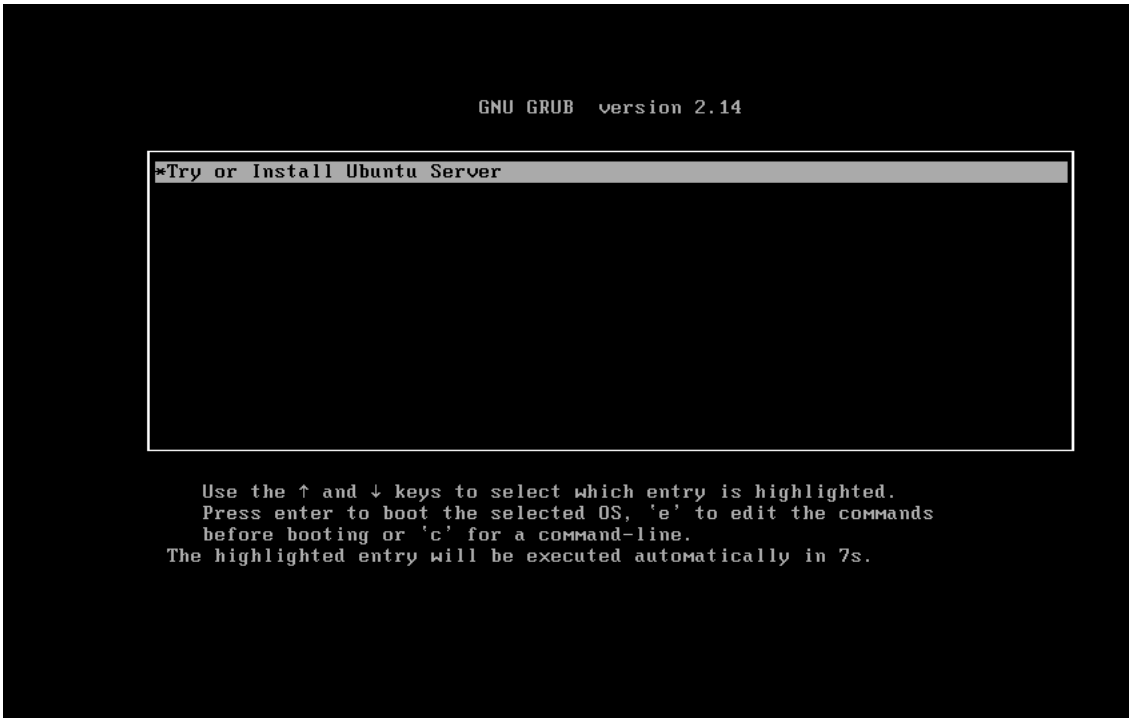


Figure 4.5:ubuntu_installer_boot.png

Caption:-Ubuntu Server Initial installer boot screen.

Task 5:-Choose the installation language:

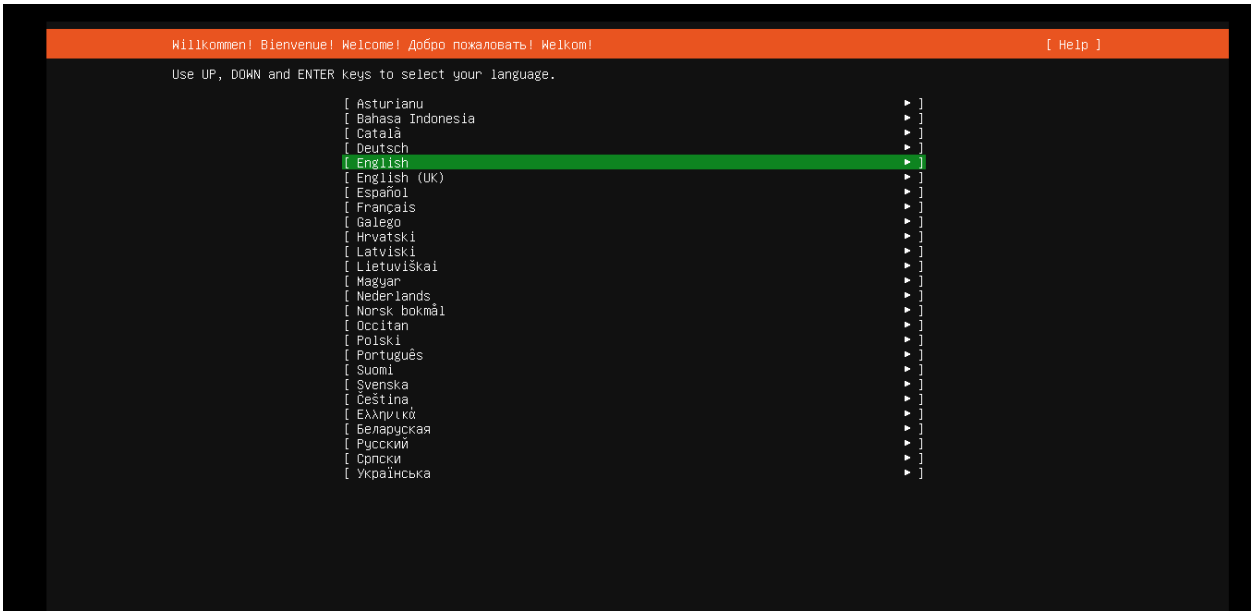


Figure 5:ubuntu_language.png

Caption:- Selecting installer language in Ubuntu Server setup.

Task 6:-Configure the keyboard layout:

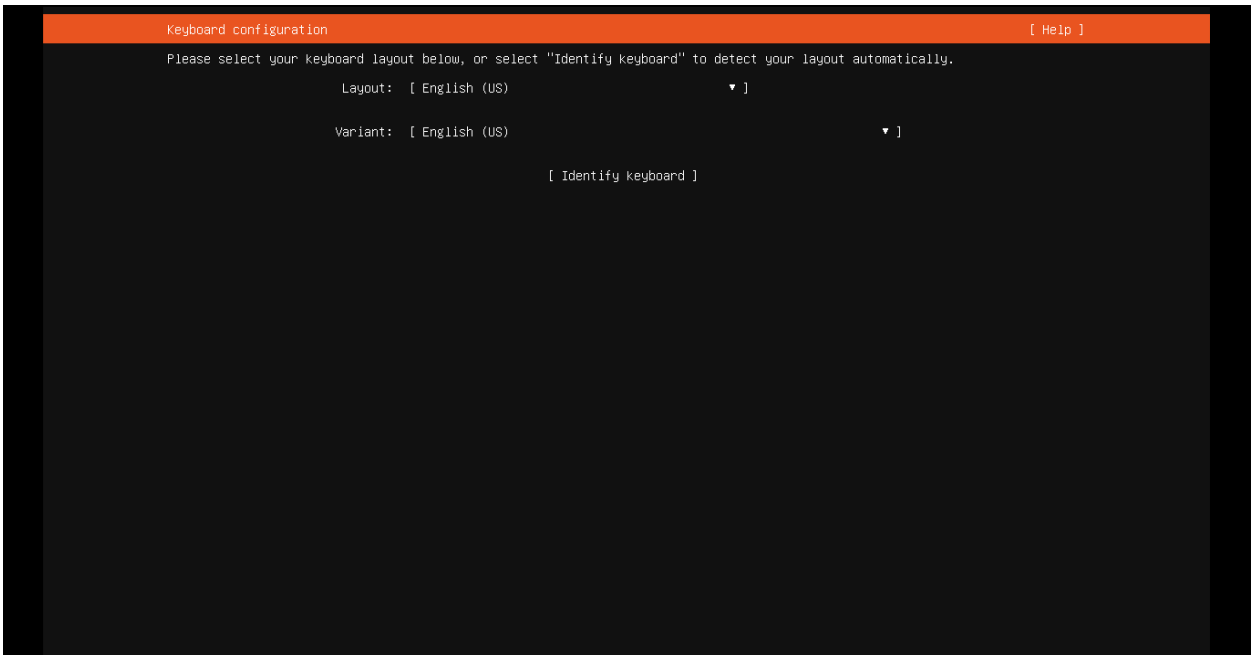


Figure 6.1:ubuntu_keyboard_layout.png

Caption: Configuring keyboard layout.

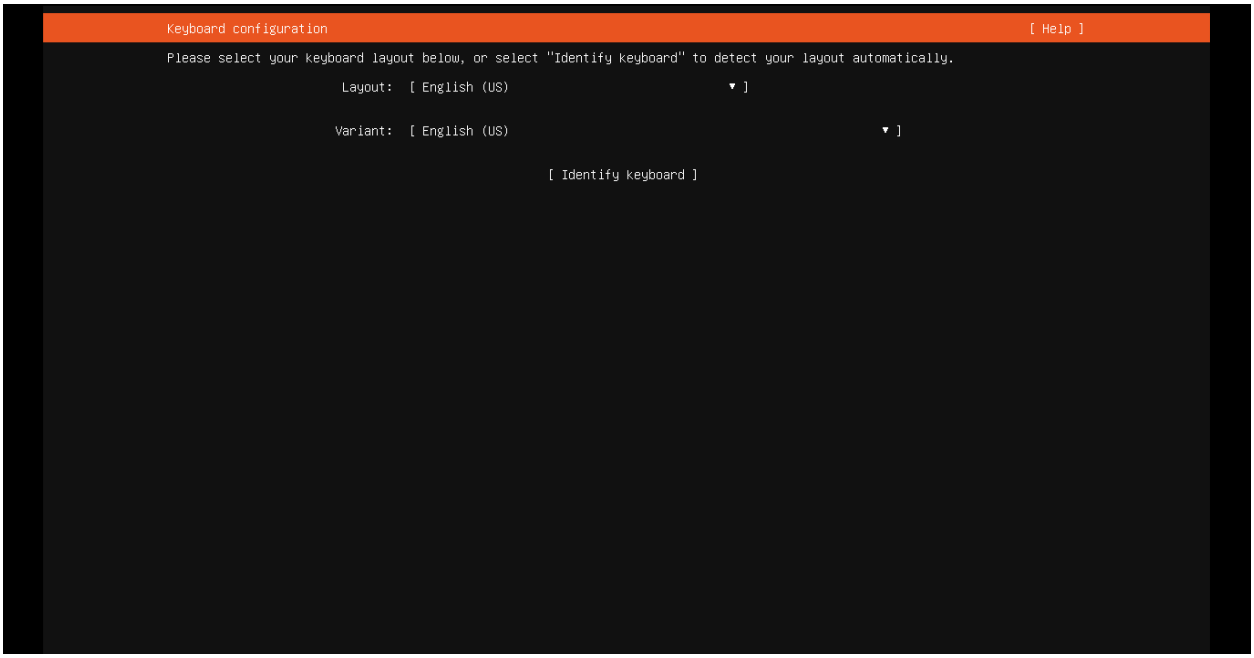


Figure 6.2:ubuntu_keyboard_variant.png

Caption:- Selecting keyboard variant settings.

Task 7:-Choose the Ubuntu Server Installation type:

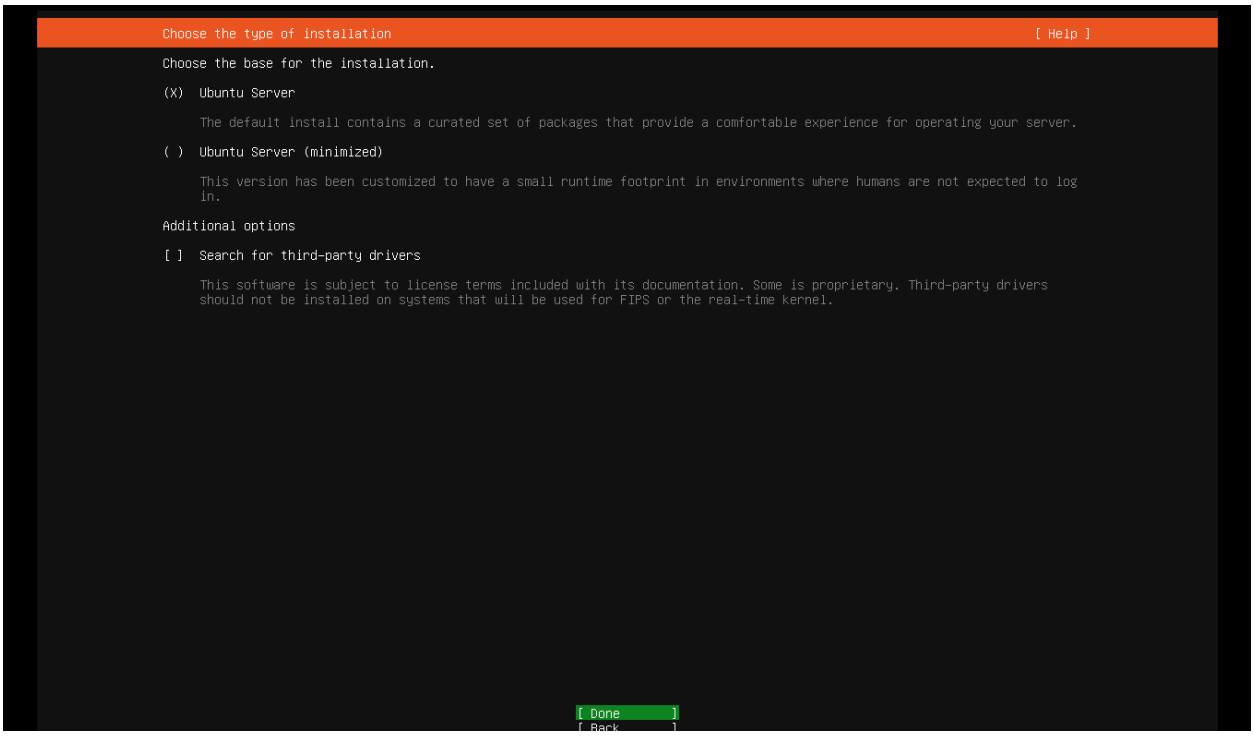


Figure 7:ubuntu_installation_type.png

Caption:-Selecting default Ubuntu Server installation option.

Task 8:-Configure the Network:

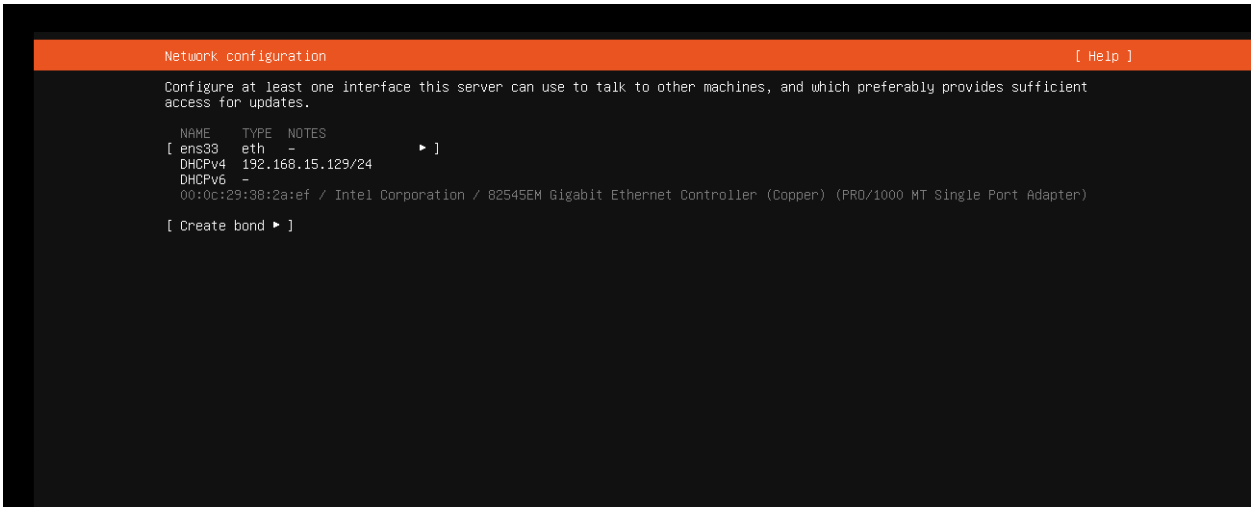


Figure 8.1:ubuntu_network_interface.png

Caption:- Detection of the network interface.

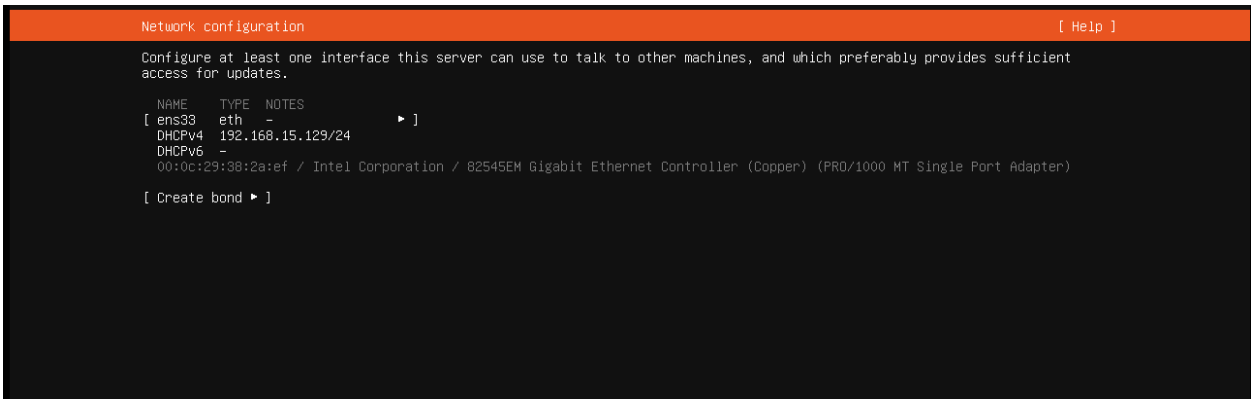


Figure 8.2:ubuntu_installer_network.png

Caption:- Successful IPv4 DHCP address assignment on the network interface.

Task 9:-Configure Storage:

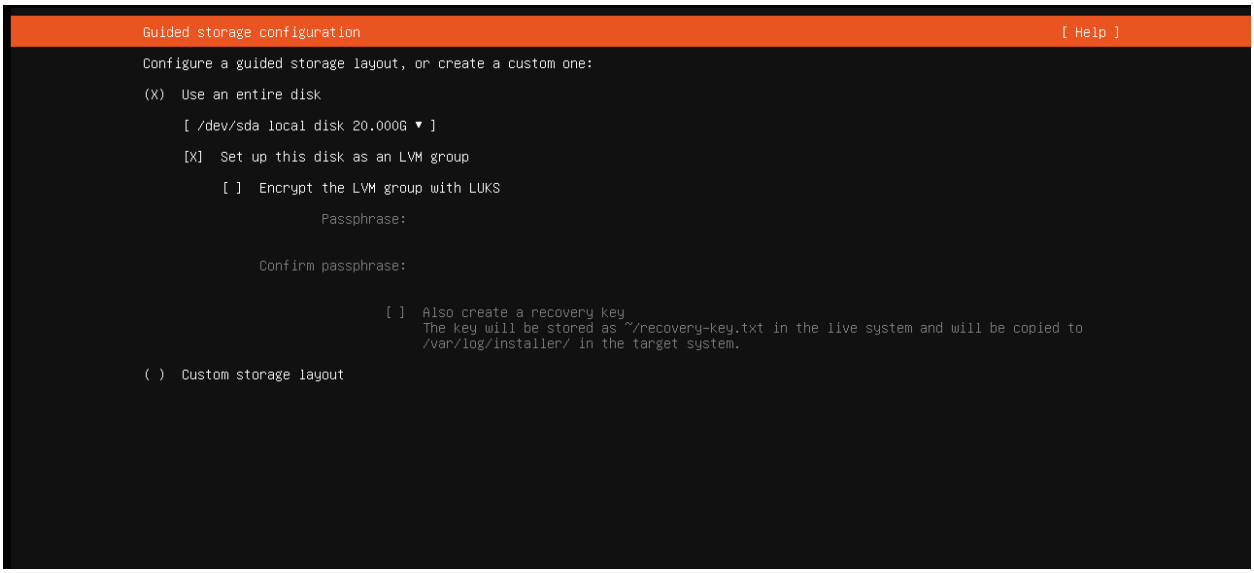


Figure 9.1:ubuntu_guided_storage.png

Caption:- Selecting guided storage setup on the VMware virtual disk.

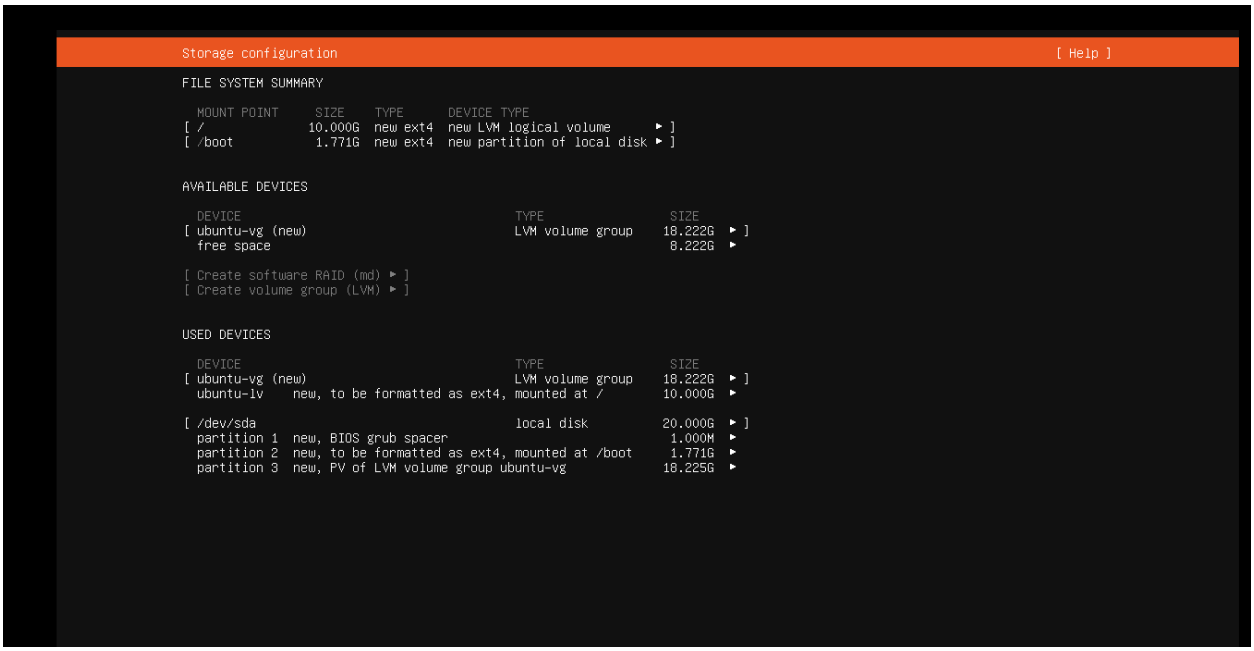


Figure 9.2:ubuntu_storage_configuration.png

Caption:- Overview of guided storage layout configuration.

Task 10:Select the Installation Disk:



Figure 10.1:ubuntu_selected_disk.png

Caption:- Selecting the VMware virtual disk for installation.

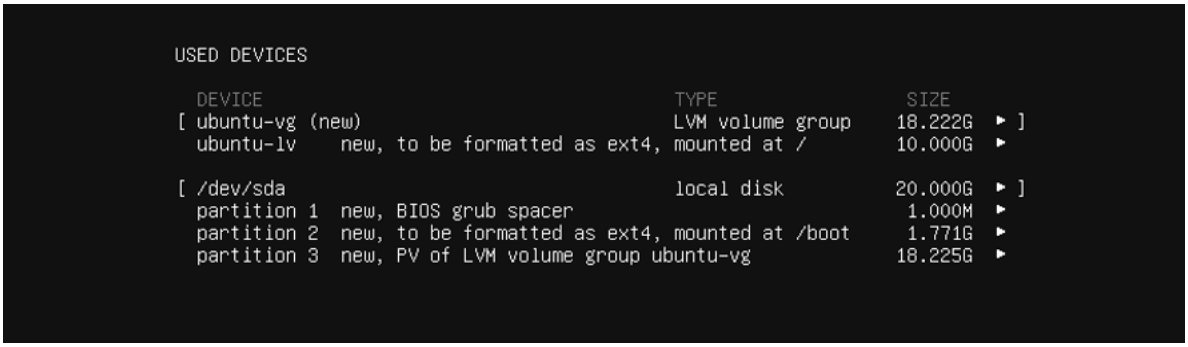


Figure 10.2:ubuntu_disk_details.png

Caption:- Verifying selected disk details and total size allocation.

Task 11:-Review the partitions:

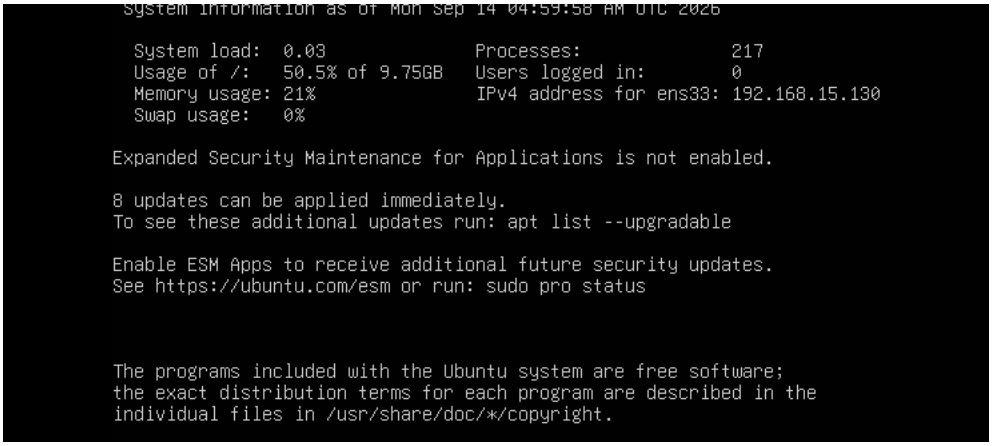


Figure 11.1:ubuntu_partition_layout.png

Caption:- Reviewing proposed filesystem partition layout on /dev/sda.

Task 12:-Confirm the Storage changes:

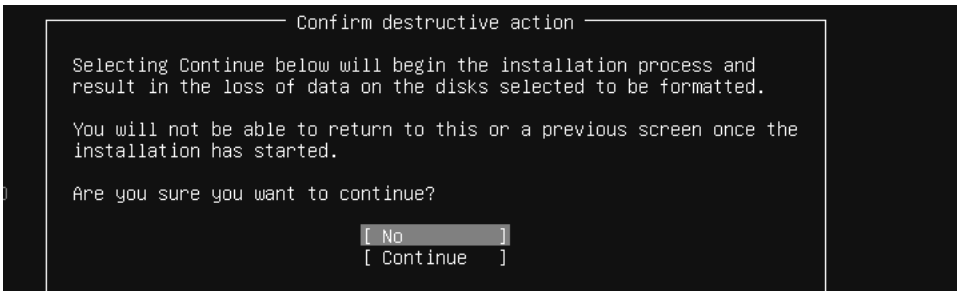


Figure 12.1:-ubuntu_storage_warning.png

Caption:- Destructive-action storage configuration warning prompt.

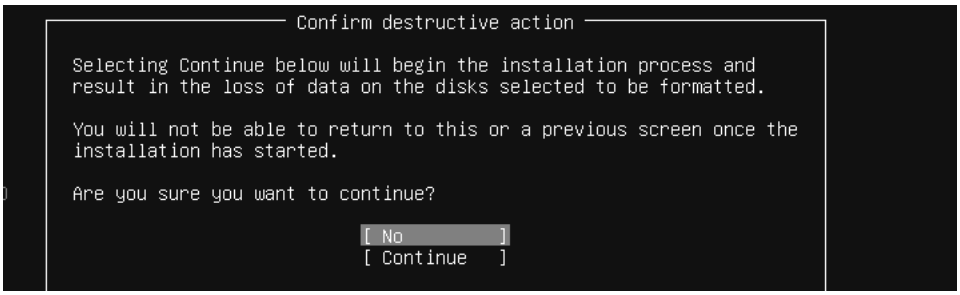


Figure 12.2:ubuntu_storage_confirmed.png

Caption:- Confirming write changes to disk.

Task 13:-Configure the Ubuntu Profile:

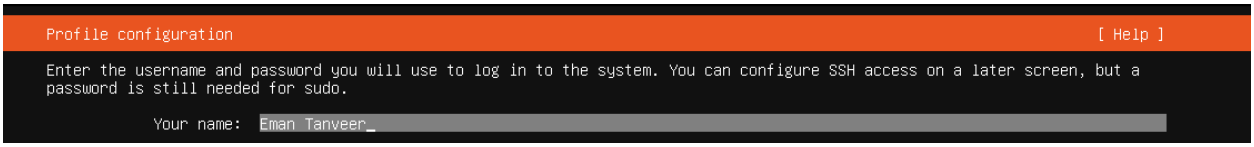


Figure 13.1:ubuntu_actual_name.png

Caption:- Entering actual full name in user profile setup.



Figure 13.2:ubuntu_server_name.png

Caption:- Configuring server hostname as ubuntu.



Figure 13.3:ubuntu_username.png

Caption:- Setting Ubuntu username to match exact GitHub username (eman-tanveer123).

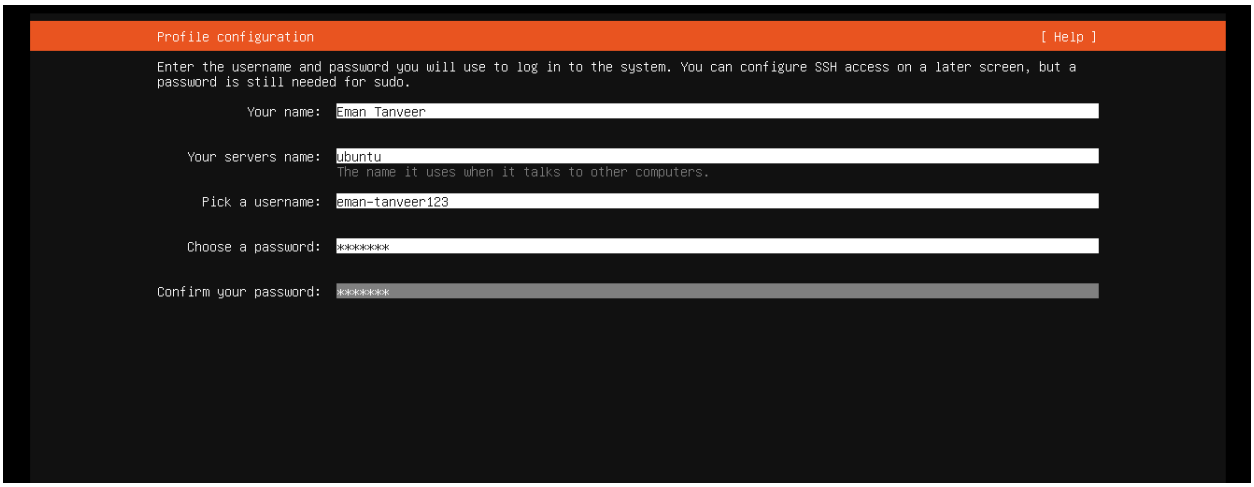


Figure 13.4:ubuntu_profile_setup.png

Caption: Completed user account and profile configuration page.

Task 14:-Complete the Software Installation:

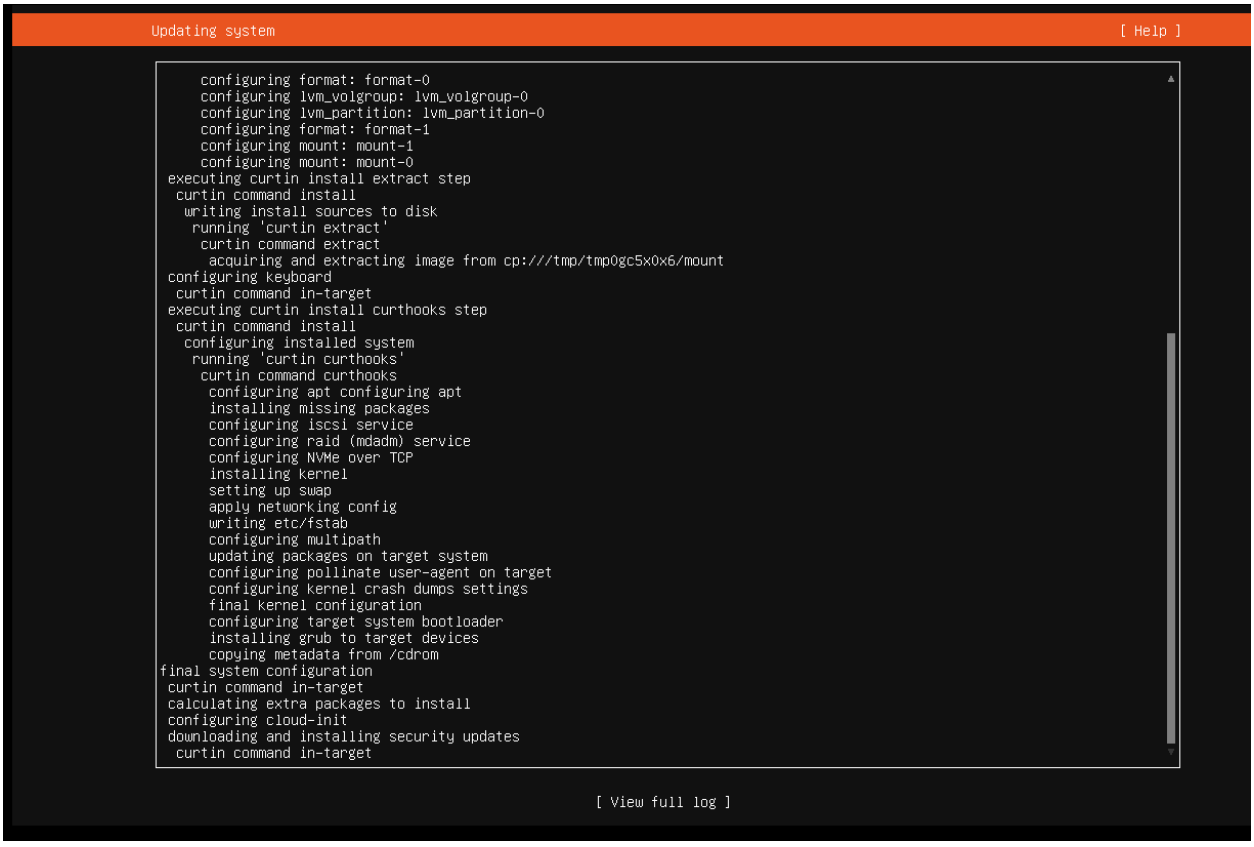


Figure 14:ubuntu_installation_progress.png

Caption:- System packages and core installation progress.

Task 15:-Reboot and Verify the Ubuntu Identity:

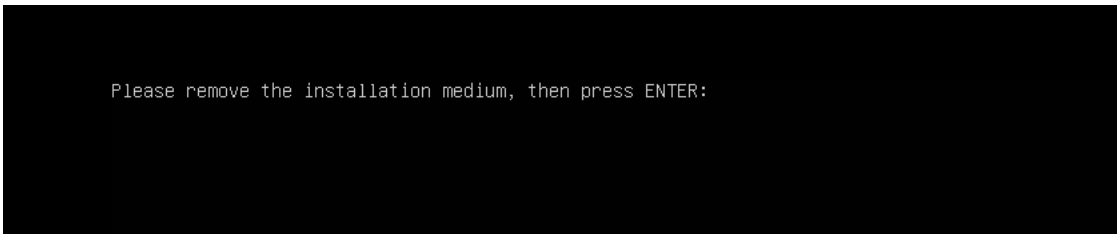


Figure 15.1:ubuntu_installation_complete.png

Caption:- Installation completed screen with reboot prompt.



Figure 15.2:ubuntu_login_screen.png

Caption:- Ubuntu Server local login prompt after initial reboot.

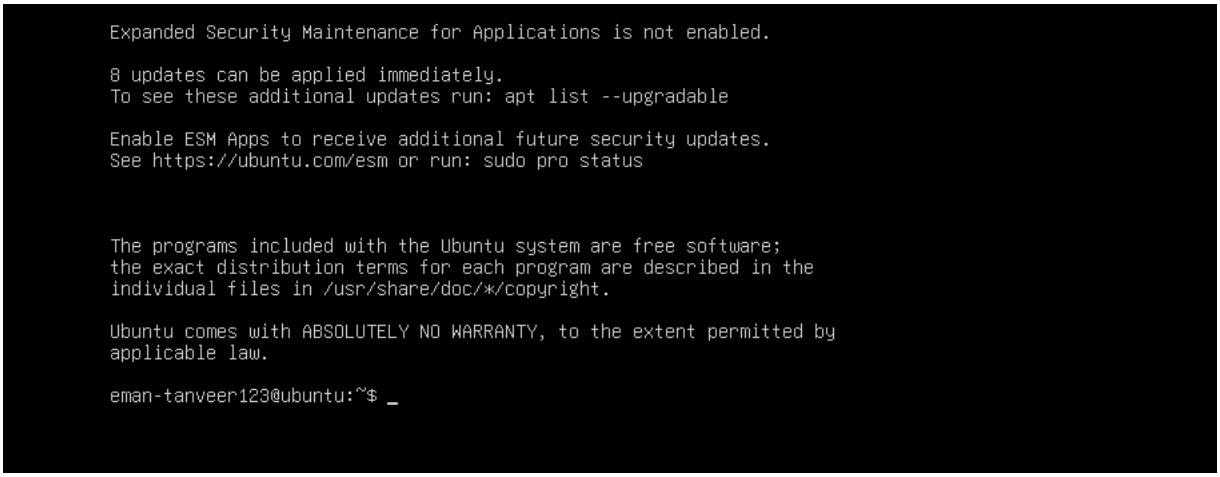


Figure 15.3:ubuntu_terminal_login.png

Caption:- Successful terminal authentication.



Figure 15.4:ubuntu_identity_verified.png

Caption:- Terminal identity prompt verification (eman-tanveer123@ubuntu:~$).

Task 16:-Find the Ubuntu Server IP Address:

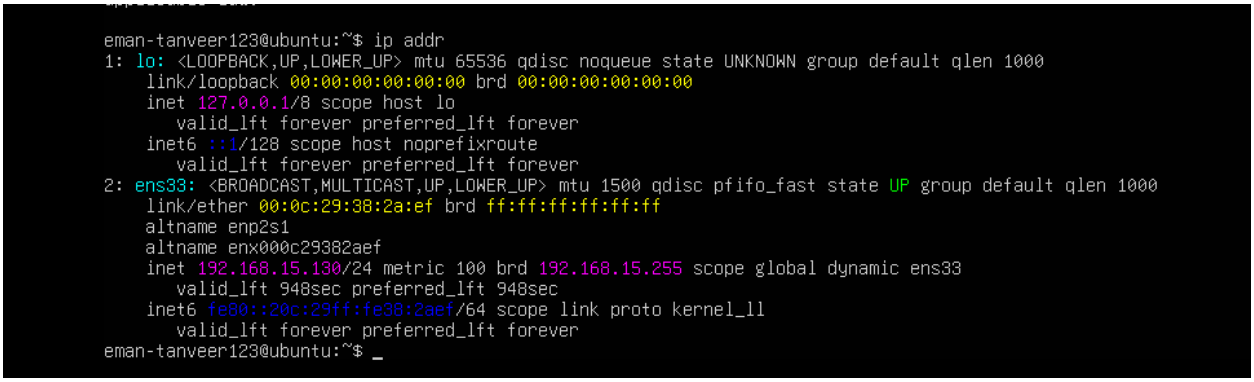


Figure 16.1:ubuntu_ip_addr_command.png

Caption:- Execution of ip addr command in Ubuntu terminal.

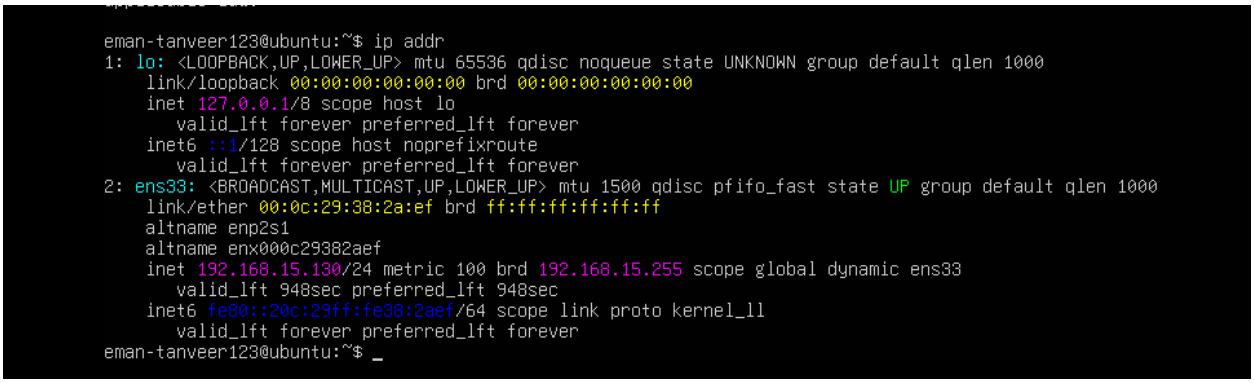


Figure 17.2:ubuntu_ip_address.png

Caption:- Displaying active IPv4 address (192.168.15.130).

Task 17:-Connect to Ubuntu Server from Windows:

```
C:\Users\nn>ssh eman-tanveer123@192.168.15.130
The authenticity of host '192.168.15.130 (192.168.15.130)' can't be established.
ED25519 key fingerprint is SHA256:exnBjT6UuDeTM8WqH1xEx25Bdqps6XiKf0rjbyp7K78.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

Figure 17.1:windows_ssh_command.png

Caption:- Executing SSH command from Windows Command Prompt (ssh eman-tanveer123@192.168.15.130)


```
C:\Users\nn>ssh eman-tanveer123@192.168.15.130
The authenticity of host '192.168.15.130 (192.168.15.130)' can't be established.
ED25519 key fingerprint is SHA256:exnBjT6UuDeTM8WqH1xEx25Bdqps6XiKf0rjbyp7K78.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```


Figure 17.2:windows_ssh_fingerprint.png

Caption:- Accepting RSA/ED25519 server fingerprint key.

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Sep 14 05:21:51 AM UTC 2026

  System load:  0.01              Processes:             226
  Usage of /:   51.0% of 9.75GB   Users logged in:       0
  Memory usage: 26%               IPv4 address for ens33: 192.168.15.130
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


eman-tanveer123@ubuntu:~$
```

Figure 17.3:windows_ssh_login.png

Caption:- Successful remote SSH authentication session.



Figure 17.4:windows_ssh_identity.png

Caption:- Remote SSH shell displaying required identity prompt (eman-tanveer123@ubuntu:~$).